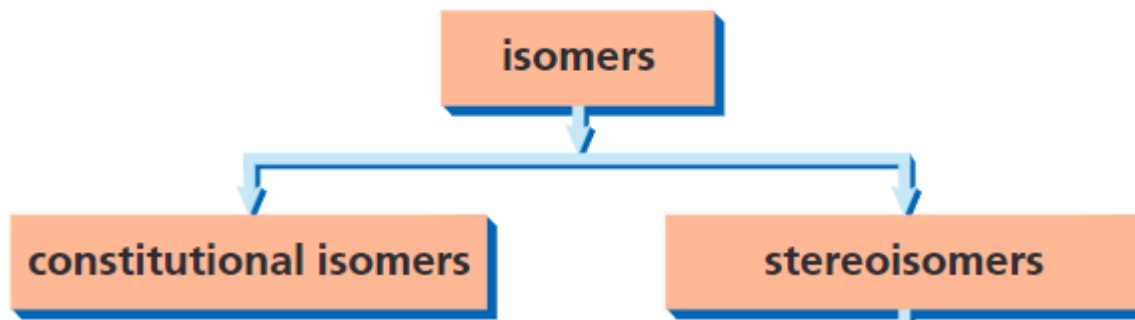


Stereochemistry:
E-Z Nomenclature & R-S Nomenclature

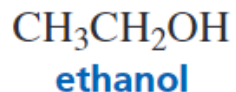
Isomers

- Compounds that have the same molecular formula but are not identical are called **isomers**.
- Isomers fall into two main classes: **constitutional isomers** and **stereoisomers**.

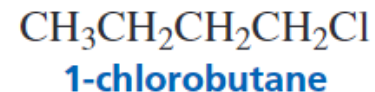
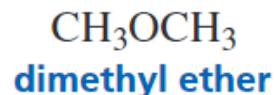


- Constitutional isomers** differ in the way their atoms are connected. For example, ethanol and dimethyl ether are constitutional isomers because they have the same molecular formula C_2H_6O , but the atoms in each compound are connected differently.

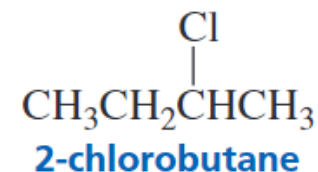
constitutional isomers



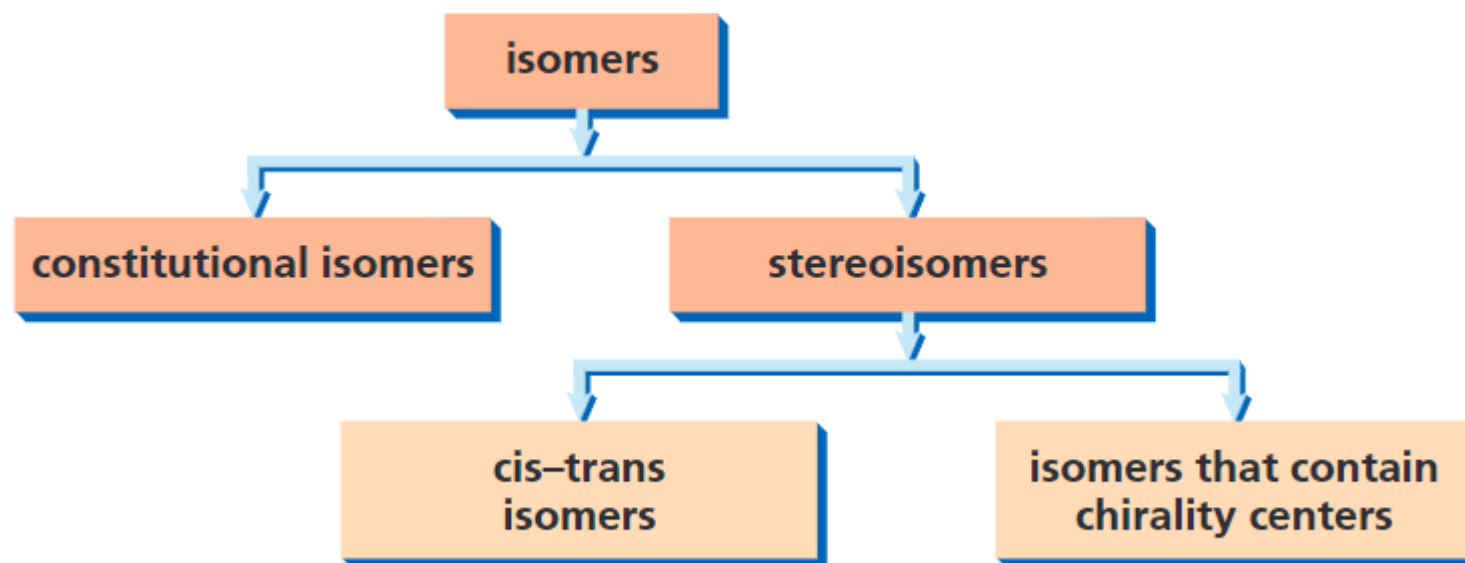
and



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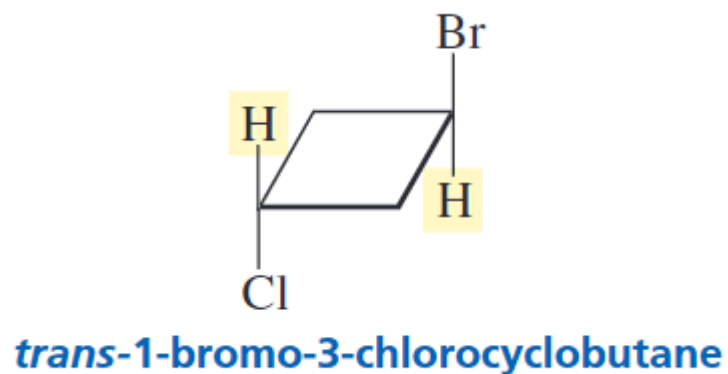
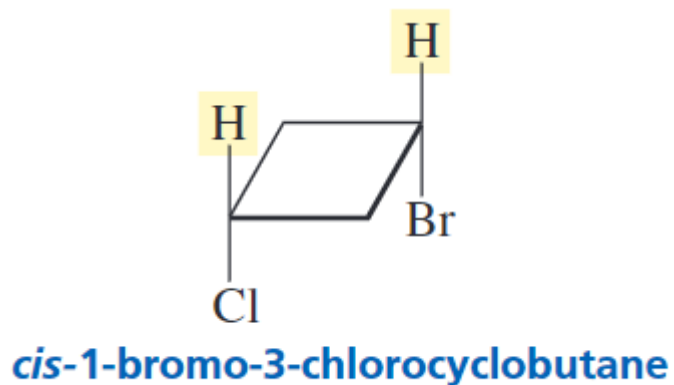
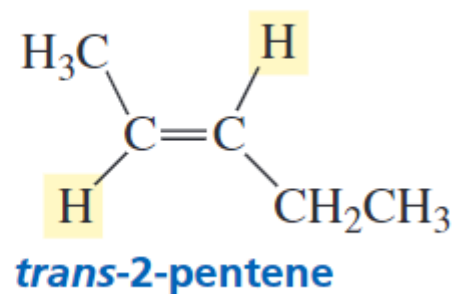
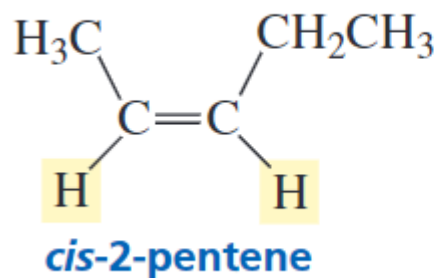


- **Stereoisomers** (also called **configurational isomers**) differ in the way their atoms are arranged in space.
- **Stereoisomers** are different compounds that do not readily interconvert. Therefore, they can be separated.
- There are two kinds of stereoisomers: **cis-trans isomers** and isomers that contain **chirality centers**.



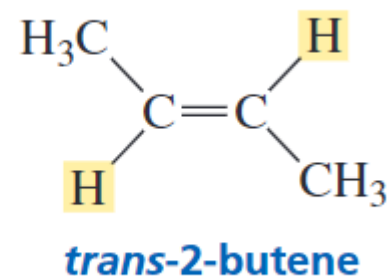
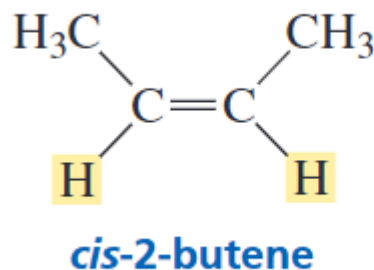
Cis-Trans Isomers

- Cis-trans isomers** (also called **geometric isomers**) result from restricted rotation. Restricted rotation can be caused either by a double bond or by a cyclic structure.

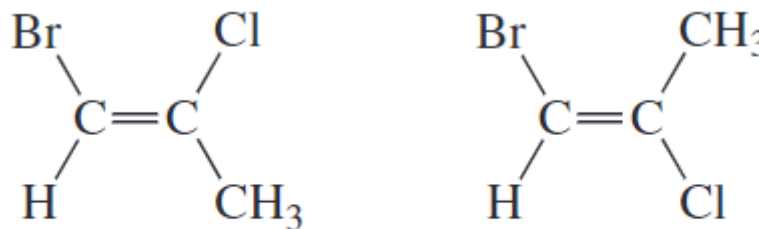


The E,Z-System of Nomenclature

- As long as each of the carbons of an alkene is bonded to only one substituent, we can use the terms **cis** and **trans** to designate the structure of the alkene.
- If the hydrogens are on the same side of the double bond, it is the **cis isomer**; if they are on opposite sides of the double bond, it is the **trans isomer**.



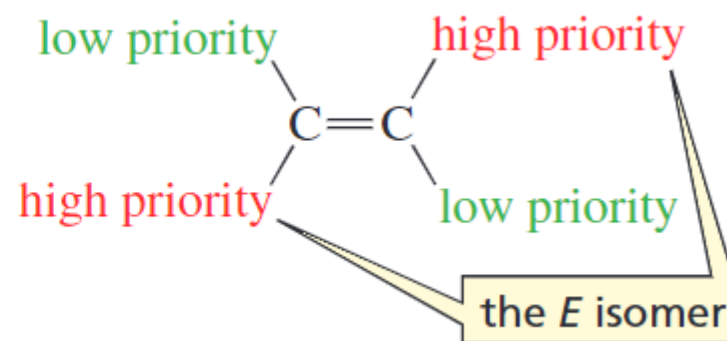
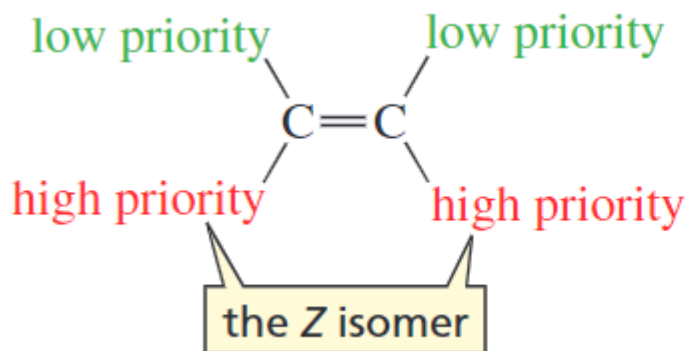
- But how would we designate the isomers of a compound such as 1-bromo-2-chloropropene?



Which isomer is cis and which is trans?

The E,Z-System of Nomenclature

- For a compound such as 1-bromo-2-chloropropene, the cis–trans system of nomenclature cannot be used because there are four different substituents on the two vinylic carbons.
- The **E,Z-system** of nomenclature was devised for these kinds of situations.
- To name an isomer by the **E,Z-system**, we first determine the **relative priorities** of the two groups bonded to one of the carbons and then the relative priorities of the two groups bonded to the other carbon.



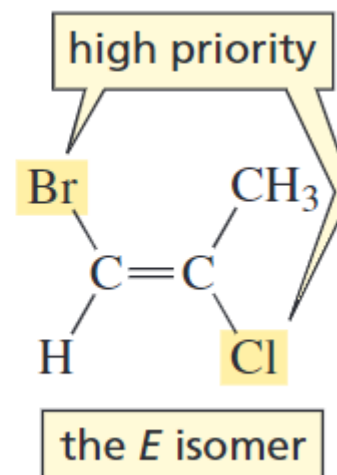
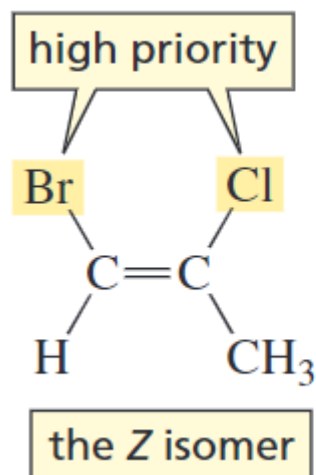
- If the high-priority groups are on the same side of the double bond, the isomer has the **Z** configuration (**Z** is for zusammen, German for “together”).
- If the high-priority groups are on opposite sides of the double bond, the isomer has the **E** configuration (**E** is for entgegen, German for “opposite”).

The E,Z-System of Nomenclature

The relative priorities of the two groups bonded to an sp^2 carbon are determined using the following rules:

- Rule 1.** The relative priorities of the two groups depend on the atomic numbers of the atoms that are bonded directly to the sp^2 carbon. The greater the atomic number, the higher is the priority.

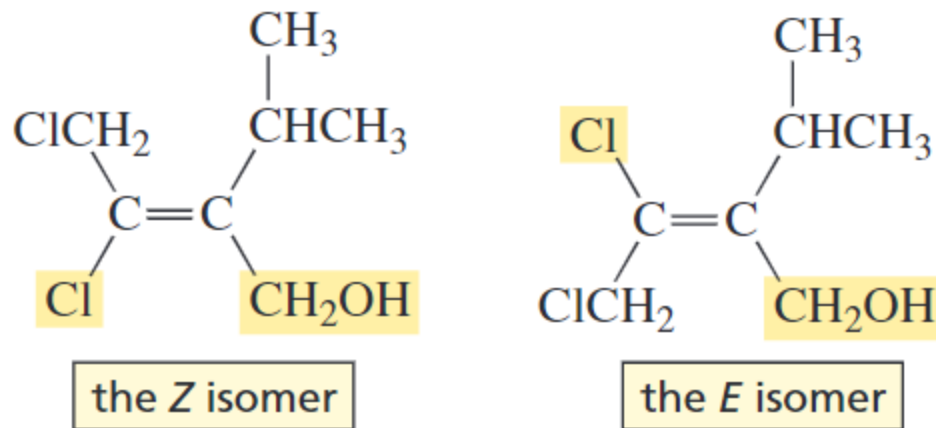
For example, in the following compounds, one of the sp^2 carbons is bonded to a **Br** and to an **H**:



The greater the atomic number of the atom bonded to an sp^2 carbon, the higher is the priority of the substituent.

The E,Z-System of Nomenclature

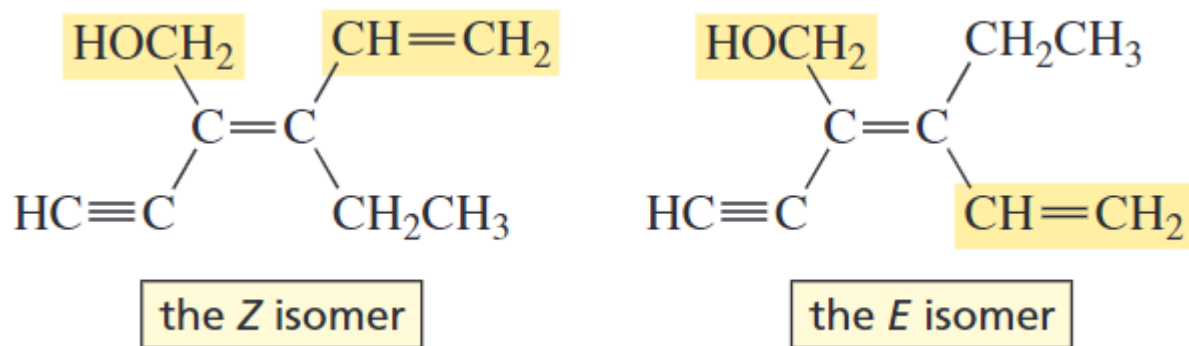
- Rule 2.** If the two substituents bonded to an sp^2 carbon start with the same atom (there is a tie), you must move outward from the point of attachment and consider the atomic numbers of the atoms that are attached to the “tied” atoms. In the following compounds, one of the carbons is bonded both to a **Cl** and to the **C** of a CH_2Cl group:



If the atoms attached to sp^2 carbons are the same, the atoms attached to the “tied” atoms are compared; the one with the greatest atomic number belongs to the group with the higher priority.

The E,Z-System of Nomenclature

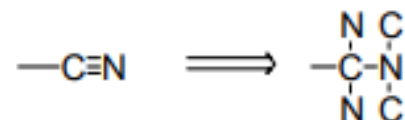
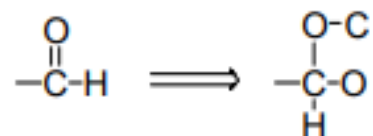
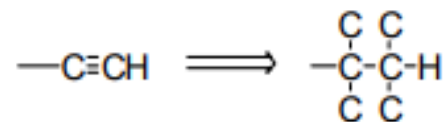
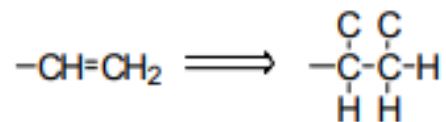
- **Rule 3.** If an atom is doubly bonded to another atom, the priority system treat sit as if it were singly bonded to two of those atoms. If an atom is triply bonded to another atom, the priority system treats it as if it were singly bonded to three of those atoms.



- If an atom is doubly bonded to another atom, treat it as if it **were singly bonded to two of those atoms**.
- If an atom is triply bonded to another atom, treat it as if it were singly bonded to three of those atoms.
- Cancel atoms that are identical in the two groups; use the remaining atoms to determine the group with the higher priority.

The E,Z-System of Nomenclature

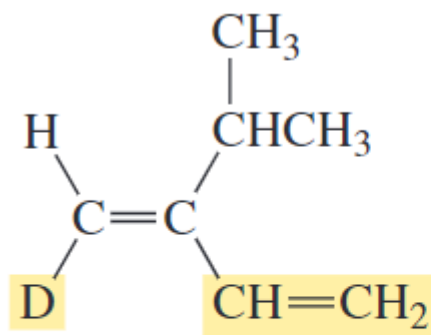
Atoms participating in double/triple bonds are considered to be bonded to an equivalent number of similar “phantom” atoms by single bonds. *Note: “phantom” atoms are bonded to no other atoms.*



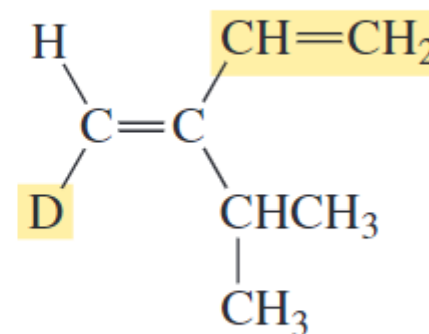
The E,Z-System of Nomenclature

- Rule 4.** In the case of isotopes (*atoms with the same atomic number, but different mass numbers*), the mass number is used to determine the relative priorities.

In the following structures, for example, one of the carbons is bonded to a **deuterium (D)** and a **hydrogen (H)**:



the Z isomer

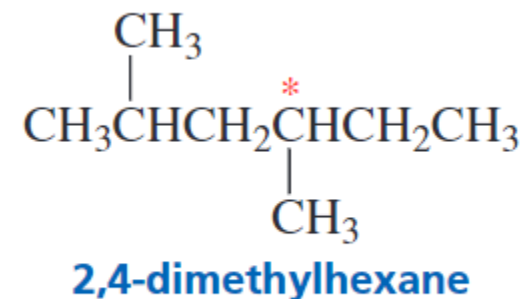
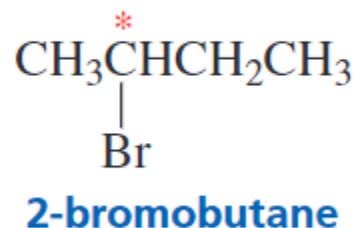
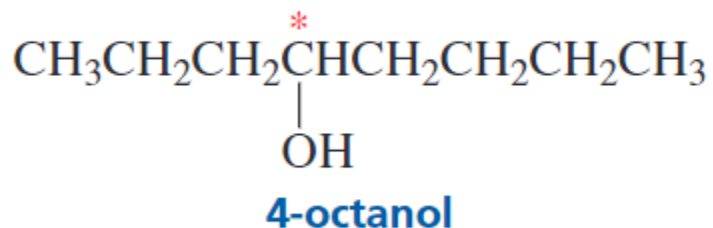


the E isomer

- If atoms have the same atomic number, but different mass numbers, the one with the greater mass number has the higher priority.

Chirality

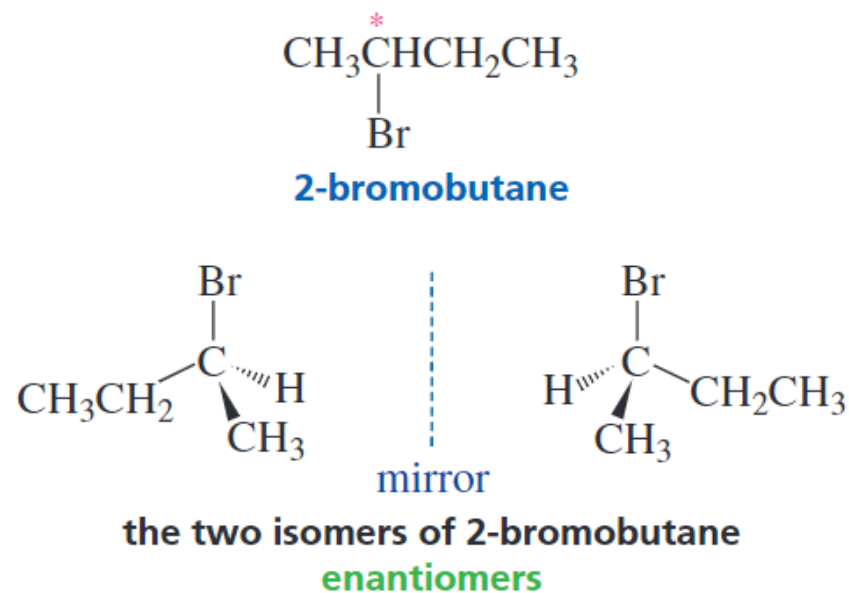
- A **chiral** object has a ***non superimposable mirror image***. In other words, its mirror image is not the same as itself.
- The feature that most often is the cause of chirality in a molecule is an ***asymmetric carbon***.
- An **asymmetric carbon** is a carbon atom that is bonded to four different groups. The asymmetric carbon in each of the following compounds is indicated by an asterisk.



- Notice that the only carbons that can be asymmetric carbons are sp^3 hybridized carbons; sp^2 and sp hybridized carbons cannot be asymmetric carbons because they cannot have four groups attached to them.
- An asymmetric carbon is also known as a **chirality center**.

Isomers with One Asymmetric Carbon

A compound with **one asymmetric carbon**, such as **2-bromobutane**, can exist as two different stereoisomers. The two isomers are analogous to a left and a right hand. Imagine a mirror between the two isomers; notice how they are mirror images of each other. The two stereoisomers are non-superimposable mirror images—**they are different molecules**.

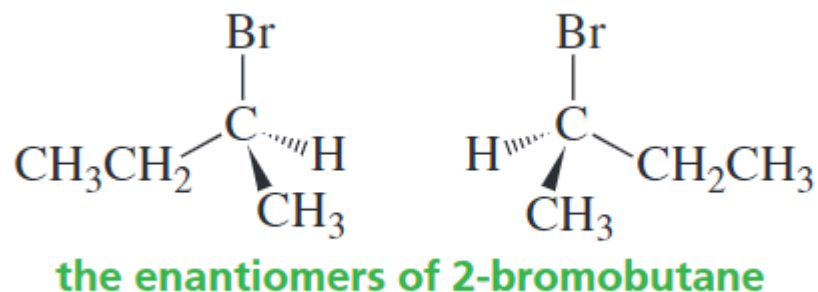


Non-superimposable mirror-image molecules are called **enantiomers** (from the Greek *enantion*, which means “opposite”). The two stereoisomers of 2-bromobutane are **enantiomers**.

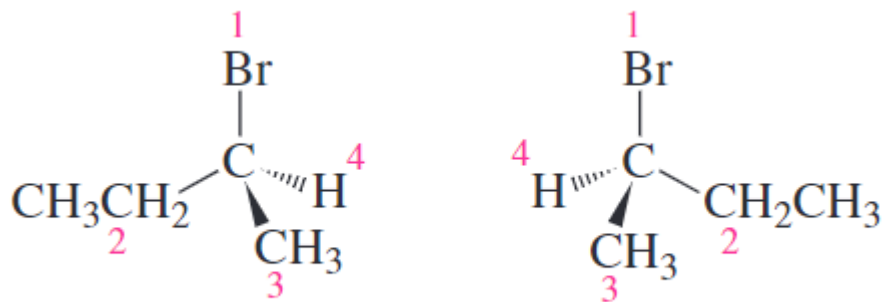
Naming Enantiomers: The R,S-System of Nomenclature

- We need a system of nomenclature that indicates the configuration(arrangement) of the atoms or groups about the asymmetric carbon.
- Chemists use the letters **R** and **S** to indicate the configuration about an **asymmetric carbon**.
- For any pair of enantiomers with one asymmetric carbon, one will have the **R configuration** and the other will have the **S configuration**.
- The **R,S-system** was devised by **Cahn, Ingold, and Prelog**.
- The letters **R** and **S** come from the Latin words '**Rectus**' and '**Sinister**' meaning 'right' and 'left'.

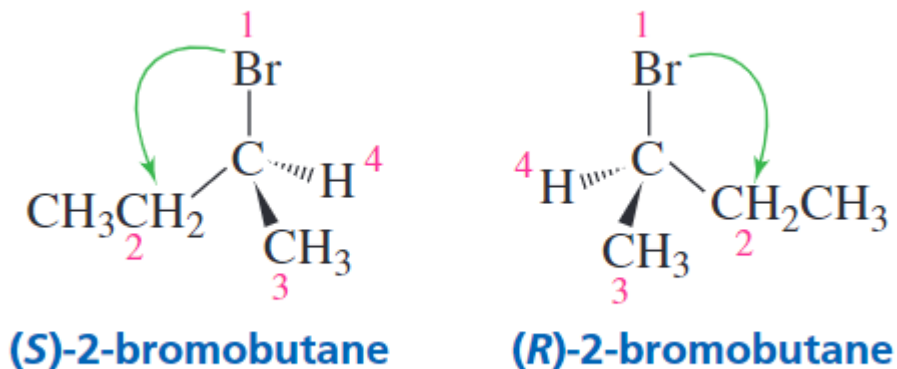
First, let's look at how we can determine the configuration of a compound drawn as a perspective formula. As an example, we will determine which of the enantiomers of **2-bromobutane** has the **R configuration** and which has the **S configuration**.



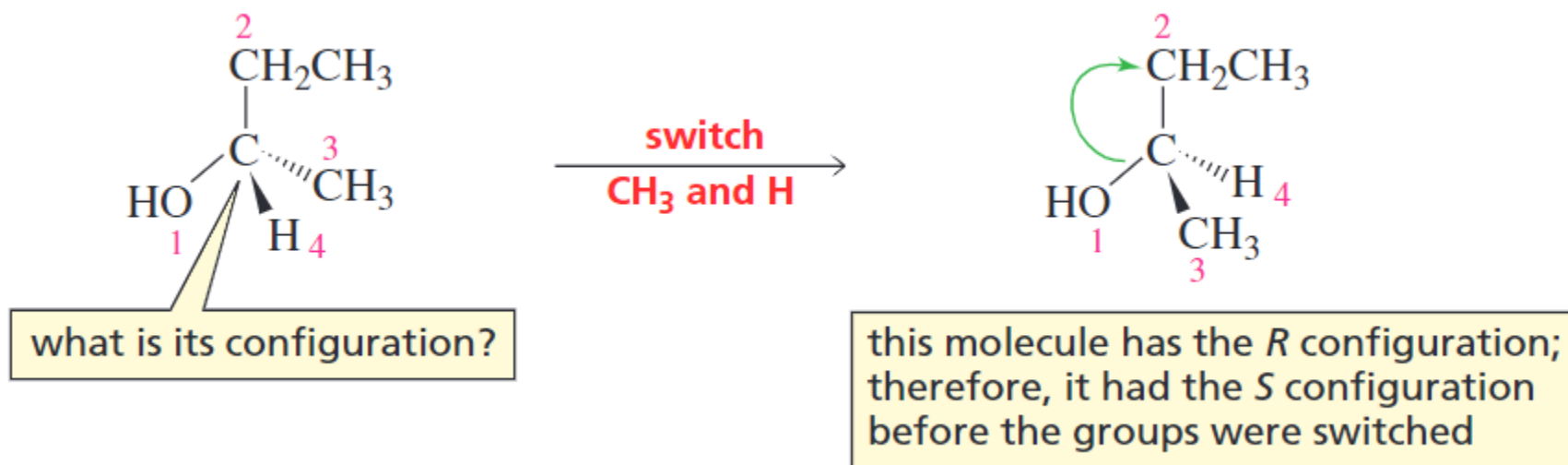
1. Rank the groups (or atoms) that are bonded to the asymmetric carbon in order of priority. In the following pair of enantiomers, bromine has the highest priority **(1)**, the ethyl group has the second highest priority **(2)**, the methyl group is next **(3)**, and hydrogen has the lowest priority **(4)**. (*priorities are assigned using the same rules as in E-Z nomenclature*)



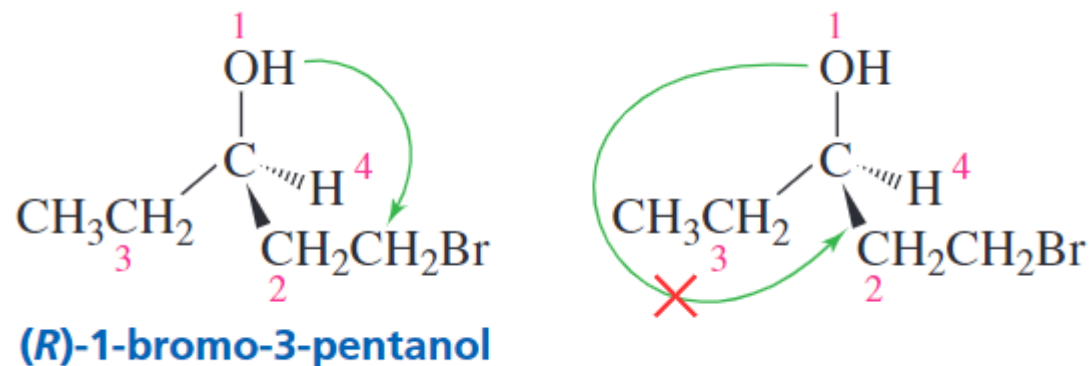
2. If the group (or atom) with the lowest priority is bonded by a hatched wedge, draw an arrow from the group (or atom) with the highest priority **(1)** to the group (or atom) with the second highest priority **(2)**. If the arrow points *clockwise*, the compound has the **R configuration**, and if it points *counterclockwise*, the compound has the **S configuration**.



3. If the group with the lowest priority (**4**) is NOT bonded by a hatched wedge, then switch two groups so group **4** is bonded by a hatched wedge. Then proceed as in step #2 (previous slide): Draw an arrow from the group (or atom) with the highest priority (**1**) to the group (or atom) with the second highest priority (**2**). Because you have switched two groups, you are now determining the configuration of the enantiomer of the original molecule. So if the arrow points *clockwise*, the enantiomer (with the switched groups) has the **R** configuration, which means the original molecule has the **S** configuration. In contrast, if the arrow points *counterclockwise*, the enantiomer (with the switched groups) has the **S** configuration, which means the original molecule has the **R** configuration.

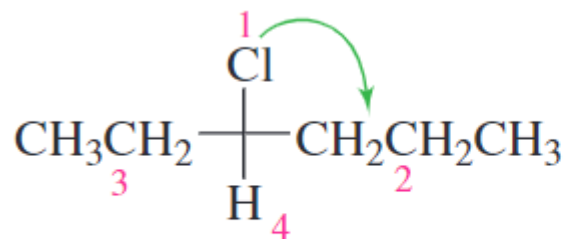


4. In drawing the arrow from group **1** to group **2**, you can draw past the group with the lowest priority (**4**), but never draw past the group with the next lowest priority (**3**).

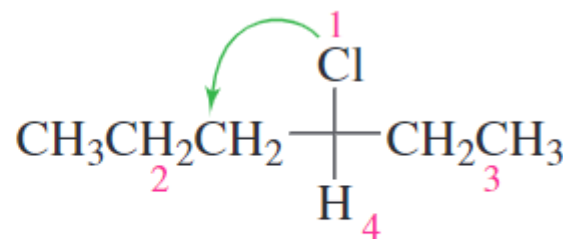


Now let's see how to determine the configuration of a compound drawn as a Fischer projection.

1. Rank the groups (or atoms) that are bonded to the asymmetric carbon in order of priority.
2. Draw an arrow from the group (or atom) with the highest priority (**1**) to the group(or atom) with the next highest priority (**2**). If the arrow points clockwise, the enantiomer has the **R configuration**; if it points counterclockwise, the enantiomer has the **S configuration**, *provided that the group with the lowest priority (**4**) is on a vertical bond*.

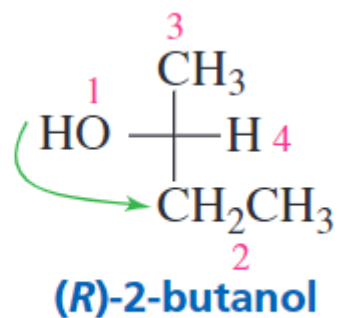
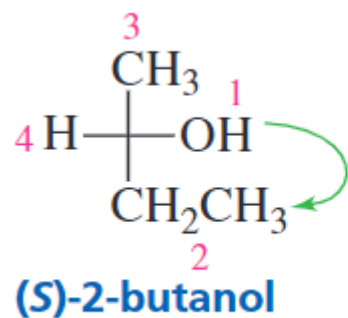


(R)-3-chlorohexane

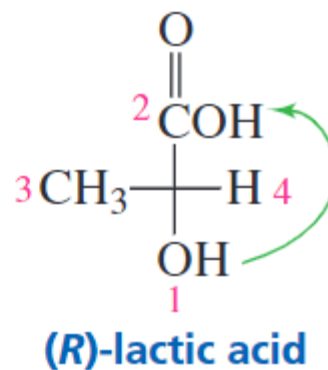
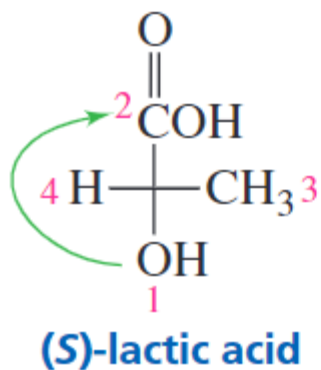


(S)-3-chlorohexane

3. If the group (or atom) with the lowest priority is on a horizontal bond, the answer we get from the direction of the arrow will be the opposite of the correct answer. For example, if the arrow points clockwise, suggesting that the asymmetric carbon has the **R configuration**, it actually has the **S configuration**; if the arrow points counterclockwise, suggesting that the asymmetric carbon has the **S configuration**, it actually has the **R configuration**. In the following example, the group with the lowest priority is on a horizontal bond, so clockwise signifies the **S configuration**, not the **R configuration**.

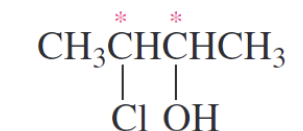


4. In drawing the arrow from group **1** to group **2**, you can draw past the group (or atom) with the lowest priority (**4**), but never draw past the group (or atom) with the next lowest priority (**3**).

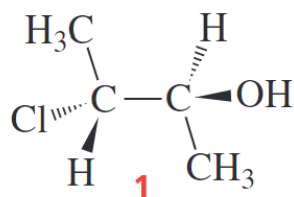


Isomers with More than One Asymmetric Carbon

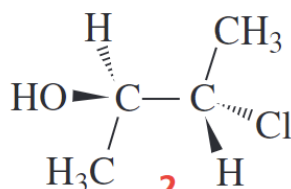
- Many organic compounds have more than one asymmetric carbon. The more asymmetric carbons a compound has, the more stereoisomers are possible for the compound.
- A compound can have a maximum of 2^n stereoisomers (provided it doesn't have any other stereocenters), where n equals the number of asymmetric carbons.
- For example, 3-chloro-2-butanol has two asymmetric carbons. Therefore, it can have as many as four ($2^n = 4$) stereoisomers.



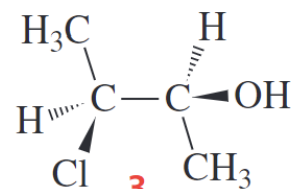
3-chloro-2-butanol



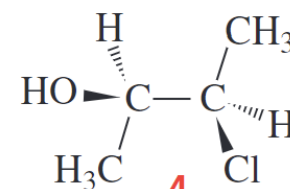
1



2



3



4

erythro enantiomers

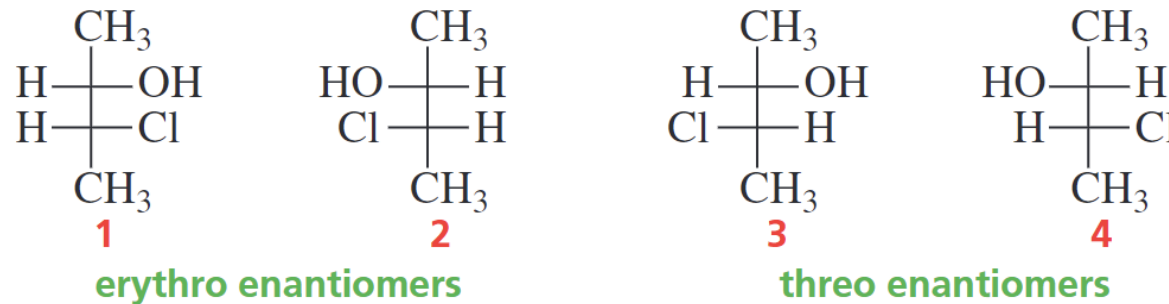
threo enantiomers

perspective formulas of the stereoisomers of 3-chloro-2-butanol (staggered)

- The four stereoisomers of 3-chloro-2-butanol consist of two pairs of enantiomers.
- Stereoisomers **1** and **2** are non-superimposable mirror images. They, therefore, are **enantiomers**.
- Stereoisomers **3** and **4** are also **enantiomers**.
- Stereoisomers **1** and **3** are not identical, and they are not mirror images. Such stereoisomers are called **diastereomers**.
- Numbers **1** and **4**, **2** and **3**, and **2** and **4** are also **diastereomers**.

Isomers with More than One Asymmetric Carbon

- **Enantiomers** have **identical physical properties** (except for the way they interact with polarized light) and **identical chemical properties**—they react at the same rate with a given achiral reagent.
- **Diastereomers** have **different physical properties** (different melting points, different boiling points, different solubilities, different specific rotations, and so on) and **different chemical properties**—they react with the same achiral reagent at different rates.
- **Cis-Trans** isomers are also considered to be **diastereomers** because they are stereoisomers that are not enantiomers.

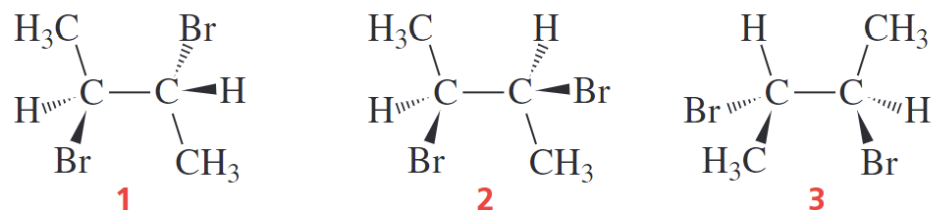
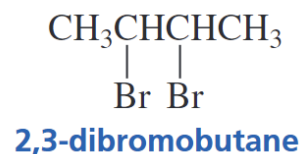


Fisher projections of the stereoisomers of 3-chloro-2-butanol

- When Fischer projections are drawn for stereoisomers with two adjacent asymmetric carbons (such as those for 3-chloro-2-butanol), the enantiomers with similar groups on the same side of the carbon chain are called the **erythro** enantiomers. Those with similar groups on opposite sides are called the **threo** enantiomers.
- Therefore, **1** and **2** are the **erythro** enantiomers of 3-chloro-2-butanol (the hydrogens are on the same side), whereas **3** and **4** are the **threo** enantiomers.

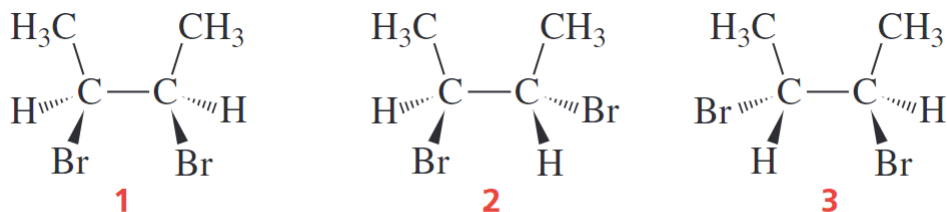
Meso Compounds

- In the examples we have just seen, each compound with two asymmetric carbons has four stereoisomers. However, some compounds with two asymmetric carbons **have only three** stereoisomers.
- An example of a compound with two asymmetric carbons that has only three stereoisomers is 2,3-dibromobutane.

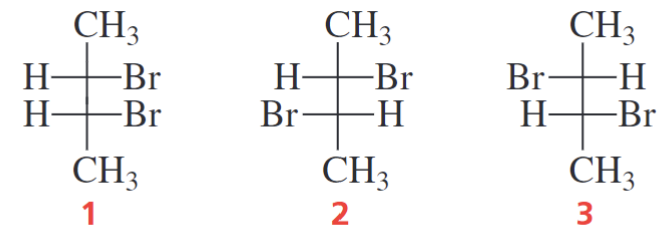


perspective formulas of the stereoisomers of 2,3-dibromobutane (staggered)

The “missing” isomer is the mirror image of **1** because **1** and its mirror image are the same molecule. This can be seen more clearly if you look either at the perspective formulas drawn in their eclipsed conformations or at the Fischer projections.

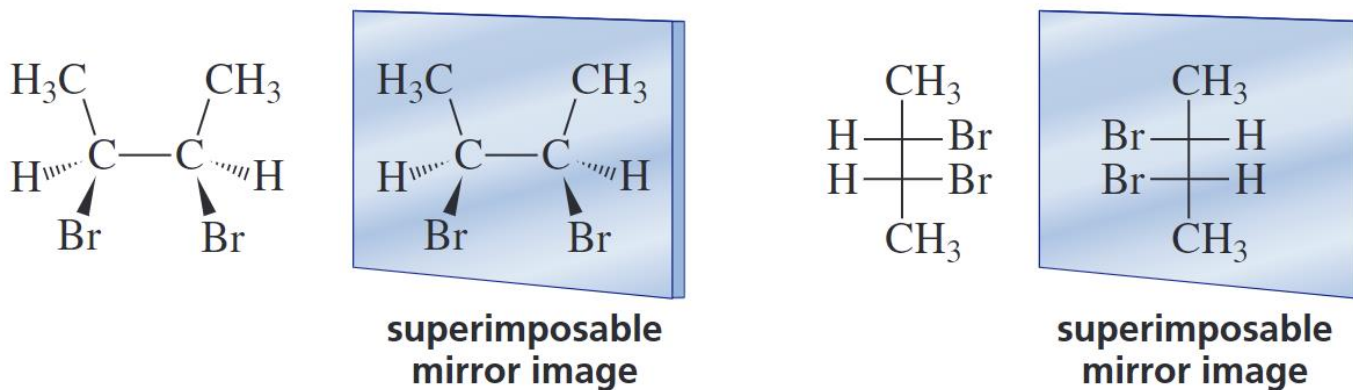


perspective formulas of the stereoisomers of 2,3-dibromobutane (eclipsed)

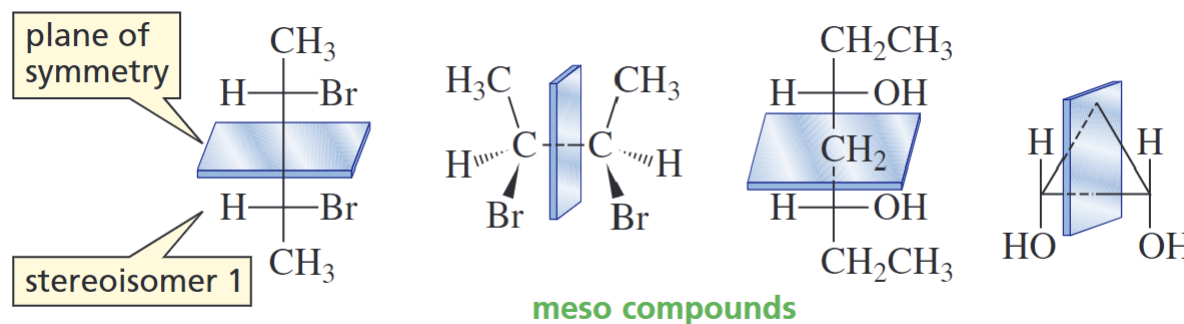


Fischer projections of the stereoisomers of 2,3-dibromobutane

Meso Compounds

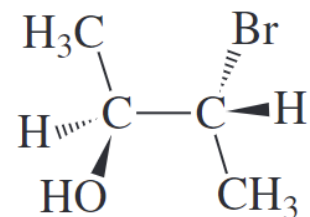


- Stereoisomer **1** is called a **meso** compound. Even though a **meso** compound has asymmetric carbons, it is an **achiral** molecule because it is superimposable on its mirror image.
- Mesos is the Greek word for “*middle*.”
- A **meso** compound can be recognized by the fact that it has two or more asymmetric carbons and a **plane of symmetry**.
- A **plane of symmetry** cuts the molecule in half, and one-half is the mirror image of the other half.
- Stereoisomer **1** has a **plane of symmetry**, which means that it does not have a nonsuperimposable mirror image—it does not have an enantiomer.



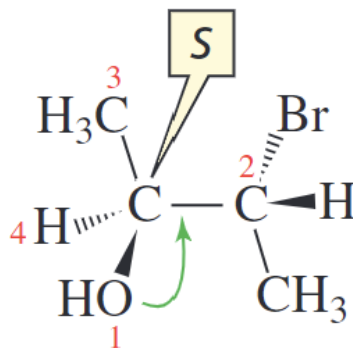
The R,S-System of Nomenclature for Isomers with More than One Asymmetric Carbon

If a compound has more than one asymmetric carbon, the steps used to determine whether an asymmetric carbon has the **R** or the **S** configuration must be applied to each of the asymmetric carbons individually. As an example, let's name one of the stereoisomers of **3-bromo-2-butanol**.



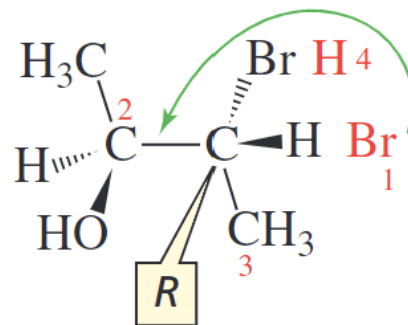
a stereoisomer of **3-bromo-2-butanol**

First, we will determine the configuration at **C-2**. The **OH** group has the highest priority, the **C-3** carbon (the C attached to Br, C, H) has the next highest priority, **CH₃** is next, and H has the lowest priority. Because the group with the lowest priority is bonded by a hatched wedge, we can immediately draw an arrow from the group with the highest priority to the group with the next highest priority. Because that arrow points **counterclockwise**, the configuration at **C-2** is **S**.



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Now we need to determine the configuration at C-3. Because the group with the lowest priority (**H**) is not bonded by a hatched wedge, we must put it there by temporarily switching two groups.



The arrow going from the highest priority group (**Br**) to the next highest priority group (the **C attached to O, C, H**) points **counterclockwise**, suggesting it has the **S** configuration. However, because we switched two groups before we drew the arrow, C-3 has the opposite configuration—it has the **R** configuration.

Thus, the isomer is named **(2S,3R)-3-bromo-2-butanol**.

